Explainer: Browsers Part 2

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Site-Specific Apps

- Some organizations with sites on the Internet also provide apps that can be used to access their sites as an alternative to accessing their sites using a browser.
- These site-specific apps are intended to give you an easier user experience than accessing the site using a browser. While these apps allow you to use the same **site** features you can use through your browser, the app's user interface has more of the "look and feel" of the device's UI. E.g., an iPhone site-specific app will provide controls for the site's features in a more "iPhone-like way" than the using the site's pages do when accessed using a browser.
- It is probably more common for an organization to provide iPhone/iPad site-specific apps than a laptop/desktop site-specific app. Although there are certainly some laptop/desktop site-specific apps out there.
- Examples of sites that provide site-specific apps:

Site	macOS app	iOS app
amazon.com	none	supported
youtube.com	various; 3rd party	supported
spotify.com	supported	supported
durhamcountylibrary.org	none	supported

What is an “AI Browser”?

- Some AI companies have created a new sub-class of browsers that are very AI-centric. I.e., these browsers have been built with agentic AI at their core.
- Per Wikipedia…
    - An AI browser is a web browser with integrated artificial intelligence capabilities, such as automatically summarizing web page content or answering questions about it. A more specialized type is an agentic browser, based on the concept of agentic AI, which can take actions – such as navigating webpages or filling out forms – on behalf of the user.
    - In the context of generative artificial intelligence, AI agents (also referred to as compound AI systems or agentic AI) are a class of intelligent agents distinguished by their ability to operate autonomously in complex environments. Agentic AI tools prioritize decision-making over content creation and do not require continuous oversight.
- Numerous Internet and cybersecurity researchers/experts warn of potential dangers using these type of browsers: They are considered to be highly vulnerable to security attacks making them easy targets for malware, fraud/scams, disinformation, and biased outputs. I recommend you **avoid using them**.
- Some of the agentic browsers currently available are ChatGPT Atlas, Perplexity Comet, Dia, Opera Neon, Fellou, Sigma
    - Note: Microsoft Edge is sometimes listed by tech experts in this sub-class of browsers; it looks to me like Microsoft has integrated their agentic AI (called “Copilot”) into the Edge browser as an additional/optional feature (meaning it is possible you could use Edge while safely avoiding it’s agentic AI capabilities.

Common Browser Features

Note:

- The number in parentheses after each feature name in the first column of the table below is the page number where detailed information on that feature begins.
- Rows highlighted with this color did not appear on the version of this slide found in the explainer handout for our first session on browsers.

Bookmarks (3)	If/when you find a page of interest to you that you to remember for use later, you can create a bookmark for that page to make it easy to access again later.
History (11)	Your browser maintains a list of pages you visit/access and a list of Internet search queries you perform. Your browser uses these lists to simplify revisiting pages and Internet searches you previously performed.
Search (12)	You can easily search for things on the Internet using the Internet search engine that is coupled with your browser. While Google Search is the most popular Internet search engine (and coupled with most browsers by default) you can change the Internet search engine your browser uses.
Cache (14)	Your browser saves some things to your local drive: cookies, browsing and search history, images, etc; browsers store these things to keep page load times short and maximize ease of use.
Built-In Password Manager (16)	Some browsers have password manager features built into them. When this feature is turned on, your browser will save the user ID and passwords of your internet accounts and use them to automatically fill in user ID and password fields when you sign into accounts.
Autofill (19)	When a page contains text boxes for you to fill in, your browser can fill in sometimes fill in these boxes for you. The kinds of text boxes in question usually have do with your personal information (name, address, phone number, etc.) How well this feature works with a given page depends in part on how the page was created.
Blocking (21)	Browsers can attempt to inhibit certain types of advertising activities (namely, ad/marketing trackers and pop-up windows); when these features are enabled, your browser will likely block some but not all pop-up windows and ad/marketing trackers; also when these features are enabled, some pages may not load or work correctly.
Extensions (24)	You can add capabilities to your browsers by installing extensions in your browser. These extensions include code that runs inside your browse, so you want to make sure any extensions you chose to install are trustworthy.
Privacy Mode (25)	You can open browser windows in this “mode.” When your browser does not “remember” (i.e., save to your local disk) the pages you access, the internet searches you do, cookies, or cached files. This feature is called “Incognito Mode” with Chrome.
Tab Groups (26)	You can create groups of pages to organize tabs you open those page in and then switch between groups.
Profiles (27)	You can create separate environments (called “profiles”) for different types of activities/work you do with your browser. For example, you can create one profile for personal activities and a second profile for professional/academic work.
Reading Mode (28)	Some browser settings/modes that remove visual distractions (like ads) on pages to help you focus on the “real” page content. This feature is sometimes also referred to as “distraction-free mode.”

Bookmarks: Overview

- Much like we use a slip of paper to save our place in a book, browser **bookmarks** provide a way to easily return to pages later without having to remember the page's URLs.
- The basic actions you can perform with bookmarks in a browser are:
    - Create a new bookmark
    - Rename an existing bookmark
    - Delete an existing bookmark
    - Organize bookmarks in folders (to help you find them easier if/when you manage to "collect" a large number of them)
- If there are some pages you visit very frequently, placing bookmarks you create for them in your browser's "**favorites**" bookmark folder can make using them even more convenient; most browsers have an app setting that will cause your browser to show you your **favorite** bookmarks in your browsers toolbar.
    - Note: The ability to show your favorite bookmarks in the browser toolbar is <u>not</u> supported on the iPhone.
- If you use the same browser on more than one of your devices, you can **sync** bookmarks across all your devices. I.e., with syncing enabled, a bookmark you create using one device will appear on your other devices. Similarly any changes you make to a bookmark on one device will on your other devices. Also, any folders you create to organize your bookmarks—along with you place inside those folders—will also appear on your other devices.

Bookmarks: Definition by other tech experts

- sigmaos.com: Bookmarks are an essential feature found in web browsers that allow users to save web pages for future reference. They are a quick and easy way to revisit your favorite websites without having to remember their web addresses. Bookmarks are also an excellent tool for keeping track of important web pages that you may need to access frequently.
- knowledgeflow.org: [Bookmarks are] a convenient way to save and quickly access your favorite websites. By creating a bookmark, you store a shortcut to a webpage, allowing you to return to it with just one click.
- biteno.com: Bookmarks are a key feature in web browsers that allow users to save and organize specific websites for easy access in the future. Instead of remembering or searching for the web address each time, you can simply click on a bookmarked link to visit your favorite sites instantly.
- laptopjudge.com: A saved shortcut (URL pointer) to a webpage for quick access in browsers like Chrome, Firefox, Edge.
- techlopedia.com: Bookmarks in web browsers are more than just links; they're gateways to important information, resources, and tools that can be revisited with ease. Creating bookmarks efficiently is the first step in building a fast and accurate digital resource library.
- wikipedia.org: In the context of the World Wide Web, a bookmark is a Uniform Resource Identifier (URI) that is stored for later retrieval in any of various storage formats.

Bookmarks: Help Pages

	Safari			Chrome
	macOS	**iPhone**	**iPad**	**Chrome**
Basics	link	link	link	link
Sync	link	*from iCloud User Guide* • Overview • "How To"		link
Favorites	link		link	

- The "Basics" pages describe how to create, delete, rename and organize bookmarks using folders
  - **Note:** bookmark folders behave similarly to file system folders but are not the same thing. I.e., you will not find any bookmark folders you create in a browser in your file manager app (i.e., Finder on Mac or Files on iPhone/iPad.)
- If you use a browser on more than one of your devices, enabling **syncing** will result in bookmarks you create on any device with that browser to also appear on your other devices.
  - Syncing must be enabled on all your devices in order for this to happen
  - Syncing with Safari is handled by iCloud's syncing capabilities.
  - For any other browser (like Chrome), you must use that browser's maker's cloud capability. I.e., in order to sync passwords you create with Chrome, you must have a Google account and be signed into that account in Chrome on all your devices.

Bookmarks: UI overview: Key Icons

Some icons that appear in browser windows are shown on the next couple of pages:

- ⋮ is called the “ellipsis” icon
- ••• is called the “meatball” icon
- ☆ is called the “favorites” icon
-  is called the “share” icon
-  is called the “sidebar” icon

Bookmarks: UI overview: Safari

- Create a bookmark
  - macOS
    - From the menu bar:
      Bookmarks > Add bookmark…
    - From the browser window:
      - > Add to “Bookmarks"
      - > Add bookmark to…
  - iOS:
    - > Add to Bookmarks (you will see “Edit Bookmark” if a bookmark already exists for the currently loaded page)
    - > Add to Bookmarks (you will see “Edit Bookmark” if a bookmark already exists for the currently loaded page)

- Access your bookmarks
  - macOS:
    - From the menu bar
      - Bookmarks > Show Bookmarks
      - Bookmarks > Edit Bookmarks
    - From the browser window: > Saved > Bookmarks
  - iPad: > Saved > Bookmarks
  - iPhone: ••• > Bookmarks
- Show favorites in the toolbar
  - macOS: from the menu bar: View > Show Favorites Bar (If favorites are already being shown, there will be a “Hide Favorites Bar” item instead)
  - iPad: Settings > Apps > Safari > General > Show Favorite Bar
  - iOS: not supported

Bookmarks: UI overview: Chrome

- Create a bookmark
  - macOS: click the found *inside* the address bar at the right hand side
    Note: The star icon will be highlighted if you already have a bookmark for the page loaded in the active/current tab
  - iPad: > Add to Bookmarks
    Note: If you already have a bookmark for the loaded page, you will see “Edit Bookmark” instead of “Add to Bookmarks”
- Access your bookmarks
  - macOS
    - From the menu bar: Bookmarks > Bookmark Manager
    - From a browser window: > Bookmarks and Lists > Bookmarks Manager
  - iOS: > Bookmarks {star icon}
- Show favorite bookmarks in the toolbar
  - macOS: > Bookmarks and Lists > Show Bookmarks Bar (also Hide Bookmarks Bar)
  - iOS: not supported

History

While using a browser, it remembers (i.e., saves to your drive) two things…

1. Pages (i.e., URLs) you access
   This saved list allows your browser to…

   - Autocomplete URLs when you type them in the browser's address bar
   - Easily return to pages you previously visited during a browsing session using the "previous page" and "next page" icons in your browser's toolbar

2. Internet search queries you enter in the browser's address bar
   This saved list allows your browser to auto-complete search queries as you type them into the browser address bar

Search

Note: It may be worthwhile for us to have a session on how you search/find things on the Internet. As this slide indicates, we have choices of Internet search services; are there pros/cons between them? Also, many sites also have search functions, and there are differences between what they do and how they do it.

- Browsers can help you find things on the Internet via it's integrated internet search capability. To use this capability, you simply type your internet search "question" (a.k.a., search **query**) in the address bar and then press the enter/return key.
- The internet search **engine** most browsers use by default is Google Search (i.e., google.com). You can change the internet search engine your browser uses via an app setting.
  - Alternate internet search engines are: Microsoft Bing, Yahoo Search, DuckDuckGo Search, Yandex, and Ecosia.

Search: Examples of “is a URL” vs “is not a URL”

When you enter a string of characters into your browser’s address bar, the browser determines if the string of characters “looks like” a URL.

1. If the browser determines the string you entered does look like a URL, the browser attempts to load the page at that URL into the view pane.
2. Otherwise, the browser sends the string to the internet search engine associated with your browser.

Here are some examples to illustrate how browsers handle various cases when you enter them in a browser’s address bar:

olli-shamlin.github.io/desig	This is a version of our SIG site’s homepage URL; its syntax is valid for a URL. If you enter this string into your browser’s address bar, your browser will load the SIG site’s home page.
wikipedia.org/mumbledeefoo	This is a URL for a Wikipedia page. Its syntax is value for a URL; however, there is no Wikipedia page named “mumbledeefoo”. If you enter this string into your browser’s address bar, you will get a “page not found” message in your browser’s view pane.
amazon.com	This is valid URL syntax that simply/only names a site; in this case the site is Amazon. If you enter this string into your browser’s address bar, Amazon’s homepage will be loaded into your browser’s view pane.
amazon	This is **not** valid URL syntax. If you enter this string into your browser’s address bar, your browser will send it to the internet search engine your browser is configured to use.
flippityfloppity.com	This is valid URL syntax that simply/only names a site. If you enter this string into your browser’s address bar, your browser will attempt to load the homepage for “flippityfloppity.com". There is no site on the Internet with that hostname, so you will get a “site note found” message in your browser’s view pane; exactly what you see in the view pane may vary across browsers.

Cache

- Your browser save some types of things to your drive as you visit sites & pages
- These “types of things” are
  - URLs of pages you visit/access
    This is called the **browsing history**
  - Internet search queries you enter into your browser’s address bar
    This is called the **search history**
  - Cookies
  - Files included in pages you visit
    Saving files included in pages can reduce the amount of time it takes pages to load
    Image files are a quintessential example

Clearing your browser’s cache

- You will find controls that allow you to clear the contents of your browser’s cache in your browser’s setting.
    - These controls tend to be found in the “Privacy” group of your browser’s settings
    - Different browsers provide varying degrees of granularity when clearing the cache. Some browsers provide very granular controls; e.g., allowing you to only delete cookies for individual sites; other browsers only give you the ability to delete cached data for a site. All browser give you the ability to delete everything in the cache at once.
- How often should you clear your browser’s cache?
    - The general “best practice” recommendation is to clear your browser’s cache once a month. If you use your browser frequently (e.g., nearly daily), consider clearing your cache every week or two. If you use your browser a few times per week and/or normally visit the same—and relatively few (say less than 20) sites—then clearing your browser’s cache 2-4 times per year may be sufficient.
    - It is possible for cached files to become “out of date” which can lead to problems loading pages; it is also possible for the cache to become so large that it can “bog down” your browser. If you ever experience any of the following when using your browser, it is worth clearing files cached for the site that is giving you trouble.
        - A page’s content appears incomplete/inaccurate
        - A page doesn’t finish loading or doesn’t load properly (i.e., it appears “hung”/“stuck”)
        - Your browser “feels” sluggish

Built-In Password Managers

- Most of the popular browsers have a password manager feature. When enabled, the browser will save the user ID and password for an account when you sign in to that account for the first time with the browser; after that, your browser will automatically fill in the user ID and password textboxes when you sign in to that account at a latter time.
- Browsers that support this feature also usually have a “sync” feature you can use to make the user IDs and passwords your browser saves available on your other devices you use the browser on. Utilizing this “sync” feature requires you to have an account for the company/organization that owns the browser.
- Note
    - Builtin-in password managers may not “recognize” the user ID and password textboxes on all sites; i.e., your browser may not autofill these textboxes on all sites when you use your browser’s password manager
    - Some browsers’ built-in password managers are more secure than others. E.g., Apple’s built-in password managers (i.e., the Passwords app) is considered highly secure while Chrome’s built-in password manager has a history of security weaknesses.

Built-In Password Managers: Help Guides

- Chrome

- Safari
  Safari does not have a built-in password manager; instead Safari integrates with Apple's Password app to get your account credentials and then auto-fill sign-in fields (e.g., user ID and password). See Apple's Passwords User Guide for more.

- Edge

- Firefox

- Opera

- Brave

- DuckDuckGo

Note: If you view this PDF using Apple's Preview app, you can click/tap on any of the underlined blue words/phrases, to access the associated browser's help information for that browser's built-in password manager.

Should I use a browser’s built-in password manager?

- In my humble opinion, it is only worth considering using a browser’s built-in password manager if/when…
  - You never use other browsers or site-specific apps
  - You have a high level of trust in the security/privacy protections the company that owns the browser provides
- When either of the above is not the case for you, I recommend using a stand alone (i.e., not built-in to your browser) password manager
  - Since you use Apple products, Apple’s Password app is a very good stand alone password manager to start with

Autofill: Definition

Note an interesting difference between the computer hope.com and techtarget.com definitions. The former states that autofill is different than autocomplete while the latter says they are the same thing!

This is an example of the ambiguous use of tech terms. Not all tech companies/experts agree on the meaning and use of terms.

- merriam-webster.com: a software feature that automatically enters previously stored information (such as a user's name or address) into a data field (as in a spreadsheet or on a web page)

- computerhope.com: In web browsers, autofill is a feature that automatically populates form fields with previously-entered information, such as passwords, addresses, and credit card data. For this sensitive information to be stored, the autofill feature must be enabled and have appropriate permissions.

  Note: Autofill should not be confused with autocomplete, a browser feature that offers suggestions as you type text in a search box or address bar.

- techtarget.com: Autofill, also called autocomplete, is a software feature that automatically inserts previously entered personal information into web form fields for the user's convenience. Autofill is often used by browsers, word processors and password managers to reduce the effort required to provide frequently requested information. Autofill is typically saved locally on the computing device it is used on. Popular fields of information completed by autofill include name, date of birth, age, address, credit card or banking information.

Autofill: Where does a browser get the information it uses to autofill?

- Safari gets
  - Your name, address, phone number from your “card” in the Contacts app
  - Credit card info from the Wallet app
  - User ID/passwords from the Passwords app
- All other browsers keep the info they use to autofill in their own private storage area
  - For some browsers, you have to directly enter autofill info yourself. Other browsers will save it on your behalf the first time you fill in a textbox that can be autofilled; this set of browsers should ask if you want them to start saving your info for autofill <u>before</u> they start saving anything
- Most browsers allow you to turn off autofill in the browser’s app settings

Links to Autofill help pages

- Safari
  - On Mac
  - On iPhone
  - On iPad
- Chrome
- Firefox
  - Page A
  - Page B
- Opera
- Brave

Blocking

- Some sites have the ability to collect information about your browser (i.e., fingerprinting) and you behavior (i.e., what you do when you visit the site). These sites can use the information they gather to “personalize” your experience using the site; in some cases this “personalization” includes targeted advertising. Also, these kinds of site may share the information they collect on your with other companies/organizations. This type of activity by a site is called **tracking**.

- Most popular browsers include the ability to prevent—or **block**—this kind of tracking behavior. There is a good deal of variability among browsers regarding the method(s) they use to block trackers, how effectively they block trackers, and when/if they attempt to block trackers. I.e., not all browsers are created equally; some browsers may do a better job blocking trackers than others. Also, not all browsers attempt to block trackers by default; you may have to change some app settings to turn a browser’s blockers.

- Some browser options related to blocking trackers allow you to add sites that are treated as exceptions; i.e., “block this behavior on all sites except …”.

- **Note:** Brave, DuckDuckGo, and Firefox are considered the leading browser for privacy and security. Brave is frequently rated as the most secure browser; DuckDuckGo is frequently rated as providing the best privacy.

Blocking: Related Safari App Settings

Safari

macOS	iOS
Websites > Pop-up Windows > When visiting other websites ~ set to “Block and Notify”	General > Block Pop-ups
Privacy > Website tracking: Prevent cross-site tracking	Privacy & Security > Prevent Cross-Site Tracking
Privacy > Hide IP address: Hide IP address from trackers	Privacy & Security > Hide IP Address
Security > Fraudulent sites: Warn when visiting a fraudulent website	Privacy & Security > Fraudulent Website Warning
Security > Non-secure connections: Warn before connecting a website over HTTP	Privacy & Security > Not Secure Connection Warning
Advanced > Privacy: Check for Highlights	Privacy & Security > Highlights
Advanced > Privacy: Use advanced tracking and fingerprinting protection	Advanced > Advanced Tracking and Fingerprint Protection
Advanced > Privacy: Block all cookies	Advanced > Block All Cookies
Advanced > Privacy: Allow privacy-preserving measurement of ad effectiveness	Advanced > Privacy Preserving Ad Measurement
Advanced > Privacy: Allow websites to check for Apple Pay and Apple Card	Advanced > Check for Apple Pay
Security > Web content: Enable JavaScript	Advanced > JavaScript

Blocking: Related Chrome App Settings

Chrome

Group		
Privacy and security	Third-party cookies	Choose “Block third-party cookies” radio button
Privacy and security	Third-party cookies	Turn on “Send a ‘Do Not Track’ request with your browsing traffic”
Privacy and security	Ad privacy	Turn off “Ad topics”
Privacy and security	Ad privacy	Turn off “Site-suggested ads”
Privacy and security	Ad privacy	Turn off "Ad measurement”
Privacy and security	Security	Safe Browsing: choose “Enhanced protection” if your are comfortable with AI else chose “Standard protection”
Privacy and security	Security	Turn on “Always use secure connections”
Privacy and security	Security	Turn on “Warn you gif a password was compromised in a data breach”
Privacy and security	Security	Turn on “Use secure DNS”
Privacy and security	Security	Manage JavaScript optimization & security: Choose “Don’t allow sites to use JavaScript optimization”

Extensions

- **Definition:** a piece of software that adds features to your web browser. Each extension typically focuses on one function, like allowing you to print a PDF file of a web page directly from your browser.

- Extensions can do almost anything. They might enable email encryption, ad blocking, one-click password storage, spell-checking, and more. Extensions are like specialized agents working with the flow of information through your browser. They might organize your notes, protect you from hackers, or just transform how that information appears in the browser window (e.g. dark mode).

- In order to function, extensions usually need broad-sweeping permissions over your browser. Some require access to almost everything your browser sees. Everything from the sites you visit, keystrokes, even your passwords. This means a bad extension (or a poorly secured browser) can expose you and your data, and introduce **major privacy and security risks**. One particularly dangerous aspect of extensions is how they can be updated automatically. This means that a popular extension could be hijacked and updated on your device and start collecting data without you ever knowing.

- Are extensions safe to use? There's no easy answer. In general, if you're downloading well-reviewed extensions from companies that you trust, you should be safe. But the **best practice is to use as few extensions as possible**. Sure, they can be handy and fun, but you shouldn't download them willy-nilly. **Only use what you need**.

Privacy Mode

- When you access/visit pages using this feature, your browser doesn't save/retain your browsing history, search history, cookies generated, cached files for the period of time you are working in privacy mode. Anyone else who also uses the same device will not be able to see which sites you visit, internet searches you perform, any cookies created or files your browser cached while you were in privacy mode.
- Note that this feature does not hide your activity from sites you visit. I.e., it does not prevent fingerprinting or tracking technologies sites use to collect data on your online behavior/activities.
- This feature goes by various names across popular browsers. It is referred to as "incognito mode" in Chrome; DuckDuckGo refer to it as "fire windows"; Safari, Brave, Opera, and Firefox refer to it as "private windows."
- When using a browser on your laptop/desktop, you should find an item in the File menu that allows you to create "private windows".
- How you create "private windows" on your iPhone/iPad seems to vary more across browsers; of those I reviewed, the controls for creating "private windows" were "in the vicinity" of the control for creating a new window or tab

Tab Groups

User Guide References

- Safari
- Chrome
- Firefox

Note: Each browser name above is a link to that browsers help page.

- A “tab group” is a feature in browsers that allows you to organize multiple tabs into labeled groups, making it easier to access and manage related pages helping reduce “clutter” while keeping your browsing experience more organized.
- Example: I use a language learning platform (duolingo.com) to study the Japanese language. When I sit down at my laptop to do my daily DuoLingo lessons, I also like to have a Japanese-English dictionary (jisho.org) and Google Translate (translate.google.com) open in browser tabs too. By creating a tab group that includes these three site’s URLs, I can…
    - Open three tabs in a browser window for each of these URLs using (almost) one click and also close them with one click when I’m done with my daily lessons
    - Move all three tabs as a single “unit" from one browser window to another

Profiles

- You can keep your personal browsing separate from your work, school, and other browsing by creating a profile for each part of your life.
- When you use profiles, changes to your browser's settings, passwords you save in a browser's built-in profile manager, and any time you clear cached data only affect the profile you are using when you do any of these things; other profiles you create will not be affected.
- To the best of my knowledge, this feature is only available with desktop/laptop versions of browsers; mobile versions (i.e., those for iPhone and iPad) do not support profiles.

Reading Mode

- When enabled, this feature attempts to hide some of a page's content (e.g., images and menus included on a page)
- This feature is also called "distraction free mode" in some browsers' help guides
- Related user guide information can be accessed when viewing this PDF document with the Preview app by clicking the browser names listed below
    - Safari Distraction Control
    - Chrome Reading Mode
    - Firefox Reader View
    - Opera Reader Mode
        - When you have a page loaded that Opera can "declutter", a book shaped icon will appear to the right of the address bar; clicking/tapping this icon will cause Opera to switch to "Reader Mode" with that page; clicking the same icon again will exit "Reader Mode".
        - This feature does not appear to be documented in Opera's help guide.
    - Brave Speedreader
    - DuckDuckGo — not supported ?